

# Note on: *Why We Don't Discount at the "Risk-Free Rate"*

Anonymous expository note

Reading notes

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## Metadata

|            |                                                                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title      | Why We Don't Discount at the "Risk-Free Rate"                                                                                                                                        |
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## One-paragraph summary

An expository piece arguing that the textbook “discount at the risk-free rate” is a placeholder for two distinct rates: the repo rate  $r^{\text{repo}}$  of the hedge instrument, which enters the *drift* of the underlying under  $\mathbb{Q}$  (and so the forward price); and the collateral rate  $r^c$  specified in the CSA, which enters the *discount factor* applied to the derivative’s value. The pre-2008 practice of collapsing both into a single LIBOR-flat rate was operationally fine when LIBOR-OIS spreads were small; the crisis forced an explicit split. The note works through three collateral regimes (perfect CSA, no CSA, partial CSA), a worked 5-year USD swap example showing a \$570k discount-rate impact on \$100m notional, and a second channel by which repo enters discounting (non-cash collateral). It is pedagogical rather than novel; equations exist chiefly to anchor the worked examples.

## 1 Setup and notation

The note uses constant-rate examples throughout for pedagogical clarity, while stating that production systems implement the stochastic versions (eqs. 1–6 in the paper).

| Paper                  | House             | Meaning                                                     |
|------------------------|-------------------|-------------------------------------------------------------|
| $r(t)$                 | —                 | Short rate (textbook)                                       |
| $r^{\text{repo}}(t)$   | $r^{\text{repo}}$ | Repo rate on the hedge instrument; drift under $\mathbb{Q}$ |
| $r_c(s)$               | $r^c$             | CSA / collateral rate; entered into discounting             |
| $r_f(s)$               | $r^{\text{fun}}$  | Unsecured funding rate of the dealer                        |
| $r_{\text{eff}}(s)$    | —                 | Path-dependent effective rate under partial CSA             |
| $V(t)$                 | $V_t$             | Derivative value                                            |
| $S(t)$                 | $S_t$             | Underlying                                                  |
| $\sigma(t, S(t))$      | $\sigma_t$        | Local volatility                                            |
| $\lambda_C, \lambda_D$ | —                 | Default intensity of counterparty / dealer                  |
| $R_C, R_D$             | —                 | Recovery rates of counterparty / dealer                     |
| $H$                    | —                 | CSA threshold                                               |

## 2 Key equations

**General stochastic pricing formula (uncollateralised).**

$$V(t) = \mathbb{E}^{\mathbb{Q}} \left[ \exp \left( - \int_t^T r(s) \, ds \right) V(T) \mid \mathcal{F}_t \right]. \quad (1)$$

*Says:* the standard textbook formula — but only when one single rate  $r(s)$  correctly captures both the cost of carry and the discounting, which holds essentially never in modern practice.

**Underlying dynamics under  $\mathbb{Q}$  (with repo).**

$$dS(t) = r^{\text{repo}}(t) S(t) \, dt + \sigma(t, S(t)) S(t) \, dW^{\mathbb{Q}}(t). \quad (2)$$

*Says:* the drift of  $S$  under the pricing measure is the *repo* rate of  $S$ , not a universal  $r$ . This is the cost of carry of the replicating hedge.

**Collateralised pricing.**

$$V(t) = \mathbb{E}^{\mathbb{Q}} \left[ \exp \left( - \int_t^T r_c(s) \, ds \right) V(T) \mid \mathcal{F}_t \right]. \quad (3)$$

*Says:* under a CSA with cash collateral earning  $r_c$ , the discount rate is exactly  $r_c$ . This is forced by the no-arbitrage martingale condition on  $\exp(-\int_0^t r_c \, ds) V(t)$ .

**XVA components** (verbatim from paper):

$$\text{CVA} = \mathbb{E}^{\mathbb{Q}} \left[ \int_t^T e^{-\int_t^s r_c(u) \, du} \lambda_C(s) (1 - R_C) (V(s))^+ \, ds \mid \mathcal{F}_t \right], \quad (4)$$

$$\text{DVA} = \mathbb{E}^{\mathbb{Q}} \left[ \int_t^T e^{-\int_t^s r_c(u) \, du} \lambda_D(s) (1 - R_D) (-V(s))^+ \, ds \mid \mathcal{F}_t \right], \quad (5)$$

$$\text{FVA} = \mathbb{E}^{\mathbb{Q}} \left[ \int_t^T e^{-\int_t^s r_c(u) \, du} (r_f(s) - r_c(s)) V(s) \, ds \mid \mathcal{F}_t \right]. \quad (6)$$

*Says:* CVA charges expected positive exposure for counterparty default; DVA gives credit for the dealer's own default risk on negative exposures; FVA charges the spread between funding and collateral rates on the net exposure. All three discount at  $r_c$ , consistent with (3).

**Pricing PDE with separated drift and discounting.** For an equity derivative on  $S$  with cash-collateralised PV:

$$\frac{\partial V}{\partial t} + r^{\text{repo}} S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = r_c \cdot V. \quad (7)$$

*Says:*  $r^{\text{repo}}$  in the drift term,  $r_c$  on the RHS. Collapsing them to a single  $r$  (Black–Scholes) misses the post-2008 distinction.

**Forward price under repo** (equity example):

$$F = S_0 \cdot e^{r^{\text{repo}} \cdot T}. \quad (8)$$

*Says:* the implied forward uses the repo rate, not the risk-free rate. The example:  $S_0 = \pounds 100$ ,  $T = 1$ ,  $r^{\text{repo}} = 5.5\%$  gives  $F \approx \pounds 105.65$ , versus the textbook  $\pounds 105.13$  at  $r = 5\%$  — a 52p difference per share before any option valuation.

### Three regimes for the discount rate :

$$\text{Perfect CSA: } V(t) = e^{-r_c(T-t)} \mathbb{E}^{\mathbb{Q}}[V(T)|\mathcal{F}_t], \quad (9)$$

$$\text{No CSA: } V(t) = e^{-r_f(T-t)} \mathbb{E}^{\mathbb{Q}}[V(T)|\mathcal{F}_t], \quad (10)$$

$$\text{Partial CSA: } V(t) = e^{-r_{\text{eff}}(T-t)} \mathbb{E}^{\mathbb{Q}}[V(T)|\mathcal{F}_t], \quad (11)$$

with  $r_{\text{eff}}(s) = r_c(s) \mathbf{1}_{\{|V(s)| > H\}} + r_f(s) \mathbf{1}_{\{|V(s)| \leq H\}}$ . *Says:* the effective discount rate is path-dependent under partial collateralisation — which is why even constant-rate partial CSAs require simulation.

### Cross-currency CSA discrepancy.

$$V^A(t) = e^{-r_c^{\text{EUR}}(T-t)} \mathbb{E}^{\mathbb{Q}}[V(T)|\mathcal{F}_t], \quad V^B(t) = e^{-r_c^{\text{USD}}(T-t)} \mathbb{E}^{\mathbb{Q}}[V(T)|\mathcal{F}_t]. \quad (12)$$

*Says:* the same derivative has two different values under two CSAs with different collateral currencies. The cross-currency basis prices this difference.

### Discount rate under non-cash (bond) collateral.

$$r_{\text{eff}} = r_{\text{repo}}^{\text{bond}} + h, \quad (13)$$

where  $h$  is the haircut cost. *Says:* when the CSA permits bond collateral, the bank must repo the bond to convert it into spendable cash; the effective discount rate is the bond's repo rate (plus haircut), not the CSA overnight rate. Distinct channel from the drift-side repo in (2).

## 3 Derivations & commentary

### Commentary

The note's structural insight is the clean separation: *drift = repo*, *discount = collateral*. Both rates are typed incorrectly in textbook treatments under the universal symbol  $r$ . The summary table in §3.3 is the operational decision rule and is worth lifting into one's own notation:

|               | Repo $r^{\text{repo}}$                 | Collateral $r^c$       |
|---------------|----------------------------------------|------------------------|
| Role          | Drift of underlying under $\mathbb{Q}$ | Discount factor on $V$ |
| Determined by | Repo market of the hedge asset         | CSA contract           |
| Can differ?   | Yes, always                            | Yes, always            |

### Commentary

The two repo channels (§3 drift and §6.5 discount-via-non-cash-collateral) are easy to conflate. They involve different bonds: the drift channel uses the repo rate of *the hedge instrument*; the discount channel uses the repo rate of *the bond posted as collateral*. These will generally trade at different rates on the same day, especially when specialness is involved.

### Commentary

The note's §6.3 worked example (5-year receiver swap, \$100m, fixed rate 3.50%,  $r_c = 4\%$  OIS,  $r_f = 4.8\%$  unsecured) gives: PV of fixed leg under perfect CSA = \$15.54m; no CSA

= \$14.97m (−\$570k); partial CSA with £10m threshold  $\approx$  \$15.29m (−\$250k). Useful as a sanity-check anchor for any multi-curve implementation.

## 4 Validation

### Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

#### Quantities the paper defines:

- Three regime-specific present-value formulas (eqs. 9, 10, 11).
- Equity forward price under repo (eq. 8).
- Bond forward price under GC repo with coupon carry ( $F^{\text{Gilt}} = P_0 \cdot e^{(r_{GC}-c)T}$ , paper §3.2).
- CVA, DVA, FVA integrals (eqs. 4–6).
- Effective discount rate under bond collateral (eq. 13).

#### Canonical numerical cases (from worked examples in the paper):

*Case A — Equity forward, §3.1:*  $S_0 = £100$ ,  $T = 1$ ,  $r^{\text{repo}} = 5.5\%$ ,  $r = 5\%$  (textbook comparison).

*Case B — 5Y receiver swap, §6.3:* \$100m notional, fix rate 3.50% received, SOFR paid,  $r_c = 4.00\%$ ,  $r_f = 4.80\%$ , 5Y tenor.

*Case C — Bond collateral ColVA, §6.5:* £20m Gilt posted as collateral on a 10Y swap, duration  $\approx 8.5$ , GC repo 4.95% vs. SONIA 5.00%, or special repo 3.50% vs. SONIA 5.00%.

#### Expected numerical values:

| Quantity                                          | Value    | Tolerance            |
|---------------------------------------------------|----------|----------------------|
| Case A: equity forward $F$                        | £105.65  | $\pm 0.01$           |
| Case A: textbook (mis)forward                     | £105.13  | $\pm 0.01$           |
| Case B: PV fixed leg, perfect CSA                 | \$15.54m | $\pm \$0.01\text{m}$ |
| Case B: PV fixed leg, no CSA                      | \$14.97m | $\pm \$0.01\text{m}$ |
| Case B: impact vs. OIS, no CSA                    | −\$570k  | $\pm \$10\text{k}$   |
| Case B: PV fixed leg, partial CSA (\$10m thresh.) | \$15.29m | $\pm \$0.01\text{m}$ |
| Case C: ColVA, GC repo case                       | £85k     | order-of-mag.        |
| Case C: ColVA, special repo case                  | £2.55m   | order-of-mag.        |

#### Equation $\leftrightarrow$ quantity map:

| Paper eq. | Symbol           | Quantity                                      |
|-----------|------------------|-----------------------------------------------|
| (2)       | $dS(t)$          | dynamics of underlying with repo drift        |
| (3)       | $V(t)$           | present value under perfect CSA               |
| (4)–(6)   | CVA, DVA, FVA    | XVA components (all discount at $r_c$ )       |
| (7)       | —                | pricing PDE with separated drift and discount |
| (8)       | $F$              | equity forward at repo rate                   |
| (9)–(11)  | $V(t)$           | PV in three collateral regimes                |
| (12)      | $V^A, V^B$       | PVs under two different CSA currencies        |
| (13)      | $r_{\text{eff}}$ | effective discount under bond collateral      |

#### Invariants and identities to verify:

1. *No-arbitrage on collateralised process*: the process  $\exp(-\int_0^t r_c ds) V(t)$  is a  $\mathbb{Q}$ -martingale under a perfect CSA; equivalently, discounting at any rate  $\tilde{r} \neq r_c$  produces a deterministic drift  $(r_c - \tilde{r})V(t)$ , which is an arbitrage.
2. *Forward reconstruction (Case A)*:  $S_0 e^{r_{\text{repo}} T} = \mathcal{L}105.65$  at the canonical inputs within  $\pm 0.01$ .
3. *Discount-rate impact (Case B)*: changing the discount rate from  $r_c = 4\%$  to  $r_f = 4.8\%$  on a \$100m 5-year swap fixed leg at 3.50% moves the PV by  $\approx -\$570\text{k}$ .
4. *Limit consistency*: the partial-CSA formula (11) reduces to (9) as  $H \rightarrow 0$ , and to (10) as  $H \rightarrow \infty$ .
5. *Cross-currency CSA disagreement*: (12) produces a strict inequality  $V^A \neq V^B$  whenever  $r_c^{\text{EUR}} \neq r_c^{\text{USD}}$ , with the sign of the gap determined by the sign of  $V(T)$  and the rate gap.
6. *Specialness  $\rightarrow$  ColVA monotonicity (Case C)*: ColVA in eq. (13) is decreasing in  $r_{\text{repo}}^{\text{bond}}$  at fixed  $r_c$ . The  $\mathcal{L}85\text{k} \rightarrow \mathcal{L}2.55\text{m}$  jump between GC and special cases reflects a 145 bp rate gap on  $\mathcal{L}20\text{m}$  at duration 8.5.

## 5 Glossary of symbols (paper's notation)

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|                              |                                                              |
|------------------------------|--------------------------------------------------------------|
| $r(t)$                       | Generic short rate (textbook)                                |
| $r^{\text{repo}}(t)$         | Instantaneous repo rate process; cost of carry for the hedge |
| $r_c(s)$                     | Stochastic collateral / CSA rate                             |
| $r_f(s)$                     | Stochastic unsecured funding rate of the dealer              |
| $r_{\text{eff}}(s)$          | Path-dependent effective rate under partial CSA              |
| $V(t)$                       | Derivative value                                             |
| $S(t)$                       | Underlying                                                   |
| $\sigma(t, S(t))$            | Local volatility                                             |
| $W^{\mathbb{Q}}(t)$          | Brownian motion under risk-neutral measure $\mathbb{Q}$      |
| $\lambda_C(t), \lambda_D(t)$ | Default intensity of counterparty / dealer                   |
| $R_C, R_D$                   | Recovery rates of counterparty / dealer                      |
| $H$                          | CSA threshold (collateral floor)                             |
| $P_0, c$                     | Bond clean price, coupon (in Gilt example)                   |
| $h$                          | Haircut cost under non-cash collateral                       |
| SOFR, SONIA, €STR            | Overnight rates in USD, GBP, EUR                             |
| GC                           | General collateral (repo)                                    |
| ColVA                        | Collateral valuation adjustment                              |

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### TODO / Open questions

- Cross-check the Case B numbers (\$15.54m perfect CSA, \$14.97m uncollateralised) against a vanilla swap pricer with a hand-built OIS-only flat curve at 4% and 4.8%.
- Implement a partial-CSA simulator that reproduces the \$15.29m blended PV for the \$10m-threshold case.
- Tab. 6.5 (Case C) gives a £2.55m ColVA gap purely from collateral specialness. Reproduce on a simple 10Y swap with bond-collateral option.